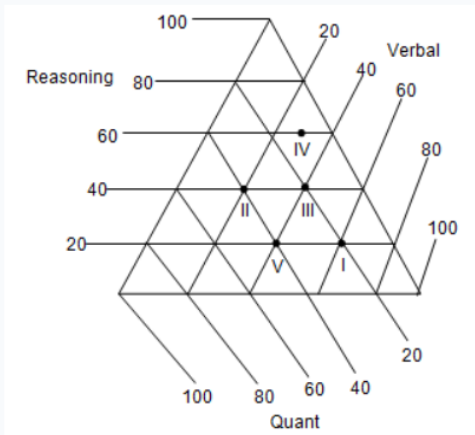


3D NETWORKS – DI_19-9-26

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

MNAT, a reputed management entrance test, allows the student a total of five attempts. The test consists of three areas – Verbal, Reasoning and Quant. The final score of the student is the sum of the average of the highest three scores in each area, in any of the five tests. For example, the verbal score could be the average of the verbal scores in test I, III and V while the reasoning score could be the average of the reasoning scores in tests II, IV and V. The following diagram gives the percentage break up of scores of a students in each of the five tests.



The table gives the total marks of the student in each of the five tests.

The table gives the total marks of the student in each of the five tests.

Test	Total Marks
I	240
II	200
III	180
IV	260
V	220

Directions for questions 1 to 5: Type in your answer in the input box provided below the question.

What is the highest score of the student in reasoning in any of the five tests?

Solution:

The scores in each area for the student for the different tests are as follows:

Test	Verbal	Quant	Reasoning	Total
I	144	48	48	240
II	40	80	80	200
III	72	36	72	180
IV	78	26	156	260
V	88	88	44	220

The highest score of the student in reasoning in any of the five tests was 156.

Ans : (156)

Directions for questions 1 to 5: Type in your answer in the input box provided below the question.

In how many of the five tests did he score more in quant than in verbal

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

The scores in each area for the student for the different tests are as follows:

Test	Verbal	Quant	Reasoning	Total
I	144	48	48	240
II	40	80	80	200
III	72	36	72	180
IV	78	26	156	260
V	88	88	44	220

The student scored more in Quant than in Verbal in only one test.

Ans : (1)

Directions for questions 1 to 5: Type in your answer in the input box provided below the question.

What is the final score of the student in reasoning (approximated to the closest integer)?

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

The scores in each area for the student for the different tests are as follows:

Test	Verbal	Quant	Reasoning	Total
I	144	48	48	240
II	40	80	80	200
III	72	36	72	180
IV	78	26	156	260
V	88	88	44	220

The final score of the student in Reasoning is

$$\frac{80 + 72 + 156}{3} = 102.67.$$

Ans: (103)

Directions for questions 1 to 5: Type in your answer in the input box provided below the question.

How many marks did the student score more in verbal in the five tests together than what he scored in reasoning?

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

The scores in each area for the student for the different tests are as follows:

Test	Verbal	Quant	Reasoning	Total
I	144	48	48	240
II	40	80	80	200
III	72	36	72	180
IV	78	26	156	260
V	88	88	44	220

Total marks in Verbal = 422

Total marks in Reasoning = 400

The difference is 22 marks.

Ans : (22)

Directions for questions 1 to 5: Type in your answer in the input box provided below the question.

What is the final score of the student?

Your Answer: Unattempted

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PREVIOUS

NEXT

Solution:

The scores in each area for the student for the different tests are as follows:

Test	Verbal	Quant	Reasoning	Total
I	144	48	48	240
II	40	80	80	200
III	72	36	72	180
IV	78	26	156	260
V	88	88	44	220

The final score of the student is

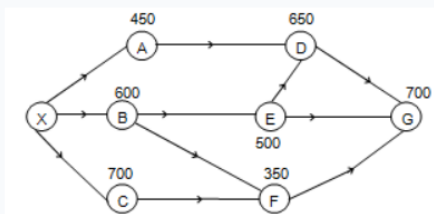
$$\frac{144 + 88 + 78}{3} + \frac{80 + 48 + 88}{3} + \frac{156 + 80 + 72}{3}$$

$$= 103.33 + 72 + 102.66 = 278.$$

Ans : (278)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The following diagram shows the network supplying oil to seven depots from a refinery X. The supply of oil is such that only after the demand at a depot is fully met, oil is passed on to the next depot in the network. The values above each depot represent the requirement at each depot and the values above the arrows represent the flow in the pipeline. The arrows represent the direction of flow of oil.



The capacity of each pipeline is 1500 kls. The slack in a pipeline is the excess flow required to bring the flow in the pipeline to full capacity

What is the minimum flow in the pipeline connecting A and X?

- ☐ a) 450 kls
- ☐ b) 950 kls
- ☐ c) 1200 kls
- ☐ d) None of these

Solution:

The total requirement at all the places is

$$450 + 650 + 600 + 500 + 700 + 350 + 700 = 3950$$

As a maximum of 1500 each can flow through X – B and X – C, 950 kl has to flow through X – A.

Choice (B)

What is the minimum slack in the pipeline connecting E and D?

- ☐ a) 850 kls
- ☐ b) 950 kls
- ☐ c) 1100 kls
- ☐ d) 1200 kls

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

If the flow in the pipeline connecting X and B is 1500 kls, after 600 kls and 500 kls are consumed at B and E respectively, 400 kls can flow through ED. As the maximum flow is 400 kls, the minimum slack is 1100 kls.

Choice (C)

What is the maximum slack in the pipeline connecting C and F?

- ☐ a) 900 kls
- ☐ b) 1050 kls
- ☐ c) 1500 kls
- ☐ d) 1250 kls

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

As seen from question 6, the minimum flow in X – A, X – B or X – C has to be at least 950 kls. After 700 kls is consumed at C, the minimum flow in the pipeline connecting C and F would be 250 kls.

∴ The maximum slack would be 1250 kls.

Choice (D)

What is the maximum flow in the pipeline connecting B and F?

- ☐ a) 900 kls
- ☐ b) 950 kls
- ☐ c) 1000 kls
- ☐ d) 400 kls

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

The maximum flow would be $1500 - (600 + 500) = 400$ kls.
Choice (D)

What is the maximum slack in all the pipelines together?

- ☐ a) 9850 kls
- ☐ b) 9650 kls
- ☐ c) 9200 kls
- ☐ d) None of these

Show Correct Answer

Hide Solution

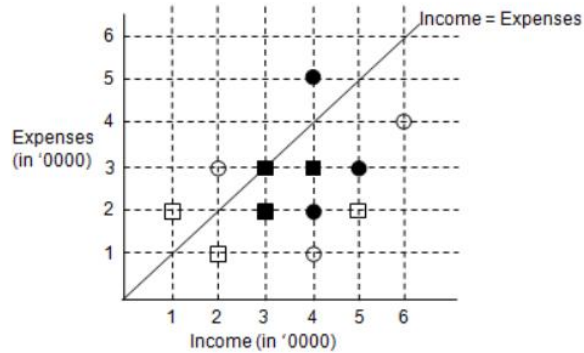
PREVIOUS

Solution:

The capacity of all the pipelines = 11×1500
= 16,500 kls.
The flow = $(450 + 600 + 700) \times 1 + (650 + 500 + 350) \times 2 + 700 \times 3$
= $1750 + 3000 + 2100 = 6850$ kls
The slack = $16,500 - 6850 = 9650$ kls
Choice (B)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The following graph gives the Income and Expenses of the different members of four families – the Raju's (□), the Iyers (■), the Kumars (○) and the Kapoors (●). Assume that the families have no other sources of income or expenditure.



Savings = Income – Expenditure

The person of which family has the highest savings?

- ☐ a) Raju
- ☐ b) Iyer
- ☐ c) Kumar
- ☐ d) More than one of the above

Solution:

The highest savings are for a member of the Kumar family.

Choice (C)

The highest value of expenditure, as a percentage of income, for any of the given persons was

- ☐ a) 200%
- ☐ b) 220%
- ☐ c) 250%
- ☐ d) None of these

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

For a member of the Raju family it is $\frac{20,000}{10,000} = 200\%$.

Choice (A)

For which family was the expenses, the highest?

- ☐ a) Raju
- ☐ b) Iyer
- ☐ c) Kumar
- ☐ d) Kapoor

Show Correct Answer

Hide Solution

PREVIOUS

Solution:

The expenses for Raju's = $2 + 1 + 2 = 50,000$

The expenses for Iyer's = $2 + 3 + 3 = 80,000$

The expenses for Kumar's = $1 + 3 + 4 = 80,000$

The expenses for Kapoor's = $5 + 3 + 2 = 100,000$

The expenses is highest for Kapoor's

Choice (D)

Which family had the highest total income?

- ☐ a) Kapoor
- ☐ b) Kumar
- ☐ c) Iyer
- ☐ d) Raju

Show Correct Answer

Hide Solution

PREVIOUS

Solution:

The income of the different families are as follow
Raju's = $1 + 2 + 5 = 80,000$
Iyer's = $3 + 3 + 4 = 100,000$
Kumar's = $2 + 4 + 6 = 120,000$
Kapoor's = $4 + 4 + 5 = 130,000$
The income is highest for Kapoor's

Choice (A)

The total savings of which family was the highest?

- ☐ a) Raju
- ☐ b) Iyer
- ☐ c) Kumar
- ☐ d) Kapoor

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PREVIOUS

Solution:

The savings are the highest for Kumar's (40,000)

Choice (C)